Exp NO: 18

K-Nearest Neighbors (KNN).

Program:

```
import numpy as np import
pandas as pd
df=pd.read_csv('Iris.csv')
df.info()
df.variety.value_counts()
df.head()

features=df.iloc[:,:-1].values label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split from
sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(
features,label,test_size=.2,random_state=1)

model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)

print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149 Data
columns (total 5 columns):
0   sepal.length   150 non-null    float64
1   sepal.width    150 non-null    float64
2   petal.length   150 non-null    float64
3   petal.width    150 non-null    float64 4   variety        150
    non-null    object dtypes: float64(4), object(1) memory
    usage: 6.0+ KB

variety Setosa
50
Versicolor    50
Virginica     50

   sepal.length  sepal.width  petal.length  petal.width variety 0
5.1          3.5           1.4           0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
```

Scores:

```
0.9583333333333334
1.0
```

Confusion Matrix:

```
[[50  0  0]
 [ 0 47  3]
 [ 0  2 48]]
```

Classification Report:

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
macro avg       0.97      0.97      0.97       150 weighted
avg       0.97      0.97      0.97       150
```